

RANTASALMI RUKKASLUOTO, Finland

WMO: 027720

Lat: 62.063N

Lon: 28.566E

Elev: 80

StdP: 100.37

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.3	-22.4	-29.2	0.3	-26.1	-24.9	0.4	-22.3	12.2	-6.2	11.1	-4.8	1.0	280	0.690

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.2	25.8	19.6	24.2	18.9	22.6	17.9	20.8	24.7	19.7	23.0	18.6	22.0	3.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	14.3	22.9	18.3	13.3	22.2	17.0	12.3	20.9	60.3	24.8	56.6	23.0	52.9	21.9	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	10.0	9.0	-28.1	27.7	4.7	2.4	-31.4	29.4	-34.2	30.8	-36.8	32.1	-40.2	33.9		
			-27.6	21.5	4.7	1.8	-31.0	22.8	-33.8	23.8	-36.5	24.8	-39.9	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.8	-9.0	-7.0	-3.6	2.3	9.6	14.7	18.3	16.8	12.0	5.4	1.0	-3.7
	DBStd	10.19	7.46	6.73	4.98	3.29	4.27	3.61	3.39	3.03	2.99	3.16	3.65	5.91
	HDD10.0	2634	589	475	423	231	59	5	0	0	16	144	269	423
	HDD18.3	5009	847	708	681	481	271	118	43	67	192	401	519	682
	CDD10.0	739	0	0	0	0	48	147	258	210	74	2	0	0
	CDD18.3	71	0	0	0	0	1	9	42	18	0	0	0	0
	CDH23.3	236	0	0	0	0	3	27	157	49	0	0	0	0
	CDH26.7	23	0	0	0	0	0	2	17	3	0	0	0	0
Wind	WSAvg	4.7	4.2	4.4	4.7	4.0	4.1	4.5	4.2	4.5	5.1	5.6	5.6	5.4
Precipitation	PrecAvg	669	50	42	36	33	45	70	84	77	59	61	60	55
	PrecMax	828	88	88	81	68	132	131	139	125	130	114	127	89
	PrecMin	512	16	5	7	12	14	22	34	24	22	17	12	20
	PrecStd	74	18	20	15	14	26	25	28	27	29	25	24	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.1	4.6	6.9	13.5	23.7	26.2	28.6	26.8	19.8	13.2	9.5	6.4
		MCWB	2.3	2.6	3.2	7.8	15.4	18.0	20.5	20.9	15.6	11.2	8.1	5.6
	2%	DB	2.1	2.7	4.9	10.7	20.7	23.6	26.5	24.8	17.9	11.7	7.7	4.3
		MCWB	1.5	1.6	2.1	6.3	14.1	17.6	20.3	19.2	15.3	10.3	6.8	3.4
	5%	DB	1.0	1.7	3.4	8.6	18.2	21.6	25.1	22.8	16.9	10.8	6.6	3.1
		MCWB	0.6	1.2	1.4	4.8	13.1	16.6	19.6	17.7	14.7	9.5	5.9	2.4
	10%	DB	0.3	0.9	2.3	6.9	16.1	20.0	23.5	21.3	15.9	9.7	5.4	2.1
		MCWB	-0.1	0.5	0.9	3.8	11.1	15.7	19.0	17.1	13.9	8.5	4.8	1.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	2.7	3.9	8.5	16.5	20.8	22.2	21.9	16.7	11.9	8.4	5.6
		MCDB	2.9	3.8	5.8	11.6	20.8	24.0	25.7	26.1	18.1	12.6	9.4	6.1
	2%	WB	1.6	1.9	2.6	6.7	15.1	18.4	21.3	19.7	15.9	10.9	6.8	3.6
		MCDB	1.9	2.6	4.3	9.9	19.6	21.8	25.2	23.0	17.3	11.4	7.5	4.0
	5%	WB	0.7	1.2	1.8	5.5	13.7	17.1	20.4	18.7	15.1	9.8	6.0	2.4
		MCDB	1.1	1.7	2.9	8.0	17.5	20.8	24.0	22.1	16.6	10.5	6.6	2.9
	10%	WB	-0.1	0.5	1.1	4.2	11.9	16.0	19.6	17.6	14.2	8.5	4.9	1.6
		MCDB	0.3	0.9	2.0	6.3	15.5	19.6	22.9	20.6	15.6	9.5	5.3	2.0
Mean Daily Temperature Range	5% DB	MDBR	5.7	5.9	7.1	6.6	6.7	5.5	5.2	4.5	3.7	3.1	2.8	3.5
		MCDBR	3.0	4.4	7.1	9.4	9.5	7.6	6.5	6.1	4.9	4.1	3.5	3.2
		MCWBR	3.1	3.9	5.3	5.4	4.6	3.9	3.2	3.2	3.1	3.4	3.4	3.1
	5% WB	MCDBR	2.9	4.4	5.6	8.3	8.6	7.1	6.2	5.5	4.4	3.9	3.6	3.1
		MCWBR	3.0	4.0	4.6	5.2	5.0	3.9	3.5	3.3	3.2	3.5	3.5	3.0
Clear-Sky Solar Irradiance	taub	0.242	0.291	0.318	0.358	0.354	0.350	0.375	0.359	0.336	0.317	0.261	N/A	
	taud	2.032	2.116	2.191	2.336	2.445	2.487	2.434	2.461	2.515	2.401	2.122	N/A	
	Ebn at Noon	425	651	782	829	867	876	840	823	779	641	400	N/A	
	Edh at Noon	37	70	94	101	98	96	99	89	70	54	33	N/A	
All-Sky Solar Radiation	RadAvg	0.22	0.78	2.28	3.65	4.90	5.31	5.10	3.91	2.21	0.87	0.23	0.10	
	RadStd	0.03	0.19	0.32	0.43	0.57	0.31	0.48	0.47	0.25	0.13	0.04	0.01	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	+0.51	N/A	N/A	N/A	N/A	N/A	-175	N/A	N/A	N/A

Nomenclature: See separate page